Toric methods for Calabi–Yau compactifications and the string landscape

Lecture notes for ESI 2026 programme on

The Unreasonable Effectiveness of Toric Geometry: Bridging Mathematics, Computation, and String Theory

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Scope and lecture outline

These lectures are an introduction to **toric methods for Calabi–Yau compactifications and the string landscape** for mathematicians. The organising question is concrete: how much of a four-dimensional effective theory, and in particular of its vacuum energy, can be computed from explicit algebraic and toric data? The candidate de Sitter construction of [1] is the central worked example, but the first goal is more basic: to explain why explicit Calabi–Yau geometry is needed at all.

The basic toric vocabulary (fans, cones, divisors, the secondary fan etc.) is used throughout without being re-derived in the live lectures; it is collected in the appendices. By contrast, the constructions that are central to the worked example, in particular Batyrev's construction of Calabi–Yau hypersurfaces from reflexive polytopes and the role of FRSTs, are re-derived at the level of a stated theorem in Lecture 1 (see Theorem 1.12). The appendices collect the definitions and standard results from algebraic geometry, convex geometry, toric geometry, mirror symmetry, orientifolds, and computational workflows that are needed to read the notes as a self-contained reference.

Sections marked by an asterisk are included for completeness and for later reference, but they are not part of the intended blackboard delivery. In the live lectures I will use them only as optional background or as answers to questions.

Prerequisites.

- Algebraic and complex geometry at the level of Huybrechts (*Complex Geometry* [2]) or Voisin (*Hodge Theory and Complex Algebraic Geometry* [3]).
- Toric geometry at the level of Cox–Little–Schenck [4]: fans, divisors, intersection rings, the Kähler and Mori cones, the secondary fan, and reflexive polytopes. Batyrev's construction [5] and the Kreuzer–Skarke list [6] are helpful background but are recalled in Lecture 1 (see Section 1.9).
- Riemannian and Hermitian geometry: holonomy and the Hodge decomposition. Berger's classification is helpful background but is restated in Lecture 1 (see Theorem 1.2).
- Basic physics vocabulary is helpful but not assumed: fields, actions, equations of motion, compactification, fluxes, branes, and effective field theory are introduced when first used.

Motivation. The motivation is two-fold. First, the notes explain how toric geometry makes certain string compactifications explicit enough that one can compute physical quantities rather than only discuss them abstractly. The conceptual ingredients have long been established: Calabi–Yau geometry, mirror symmetry, Picard–Fuchs equations, Gopakumar–Vafa invariants, D-branes, and four-dimensional supergravity have all been studied for decades. What has changed is the feasible computational range. Many calculations that were once limited to one- or few-modulus examples can now be attempted for the largest known Hodge numbers, i.e., compactifications with hundreds of moduli, provided one keeps careful control of truncations and chamber structure.

Second, the notes are meant to identify bottlenecks. The concrete questions are geometric before they are physical. Given a Calabi–Yau threefold X, how does one determine its Kähler cone, effective cone etc.? How does one compute minimal-volume representatives of divisor classes, especially when holomorphic representatives exist only after moving in the space of birational equivalence classes of X? Which effective divisors have $h^\bullet(D, \mathcal{O}_D) = (1,0,0)$ and can therefore contribute to the non-perturbative superpotential? How should one compare the Wall data of hypersurfaces arising from different triangulations, and how many inequivalent geometries are really present in the Kreuzer–Skarke ensemble? The hope is that the lectures make these questions precise enough to be useful to both the mathematics and string theory communities.

Lecture outline. The live blackboard narrative is deliberately narrower than the written notes: compactification first selects Ricci-flat internal geometries; the deformations of those geometries become four-dimensional moduli; fluxes, sources and quantum corrections generate a scalar potential V; and toric geometry supplies much of the discrete data needed to write and analyse that potential. The detailed explanation of how ten-dimensional fields reduce to four-dimensional fields appears only after the Calabi–Yau moduli have been introduced. In this sense the purpose is not to give a general survey of string theory, but to explain why explicit geometry is needed to compute its four-dimensional consequences.

There is intentional overlap with the TASI lectures [7]. The TASI notes are the physics-first reference for the Type IIB flux compactification, the leading-order effective field theory, and the candidate de Sitter construction. The present notes reorganise that material for the ESI audience: the blackboard route starts from compact Ricci-flat geometry, then explains which toric computations supply the input to the four-dimensional scalar potential. Whenever a statement is imported from the TASI treatment rather than rederived here, the relevant TASI section is cited explicitly.

1. **Lecture 1: The why and how of Calabi–Yau.** The first lecture introduces the minimal string theory input, explains why compactification leads to a Ricci-flat internal geometry, why Calabi–Yau threefolds are the computable and best understood class used in these notes, and how Batyrev's construction turns reflexive polytopes and FRSTs into explicit Calabi–Yau hypersurfaces in toric varieties. The emphasis is pragmatic: what toric methods compute reliably today (Hodge data, intersection numbers, cones, divisor topology), and what remains beyond current algorithms. Along the way we preview where these data enter physics: flux superpotentials, instanton corrections, divisor zero-mode tests, and Kähler-cone navigation.
2. **Lecture 2: Non-perturbative physics from toric data.** The second lecture specialises to Type IIB on Calabi–Yau orientifolds. Orientifolds are introduced in Lecture 2, after the unorientifolded Calabi–Yau moduli have been fixed in Lecture 1. It introduces the four-dimensional $\mathcal{N} = 1$ supergravity data only to the extent needed to compute a scalar potential: the fields (τ, z^i, T^a), the Kähler potential, the flux superpotential, the tadpole constraint, and the possible non-perturbative terms. The main point is how toric computations enter these functions: periods and genus-zero GV invariants determine W_{flux} and the small-W_0 racetrack, while divisor Hodge data screen possible ED3/gaugino-condensate terms in W_{np}. This culminates in the perturbatively flat vacuum mechanism, where modest integer flux data and exponentially small instanton terms combine to produce a very small flux superpotential.
3. **Lecture 3: Vacuum energies from toric geometry.** The third lecture assembles the leading-order effective theory for Type IIB O3/O7 orientifold compactifications with flux, non-perturbative Kähler-modulus stabilisation, and a redshifted antibrane uplift. It introduces geometric phases, flop transitions, vex chambers, and the extended Kähler cone, then explains how chamber-aware algorithms, including `fanroots` and `JAXVacua`, locate Kähler moduli critical points. The remaining ingredients are the Klebanov–Strassler throat, KPV anti-D3 uplift, the explicit toric AdS precursor and candidate-dS recipes, and the control checks: tadpoles, GV convergence, higher-order corrections, backreaction, and quantisation issues. The list is retained because these checks distinguish the leading-order candidate construction from a fully controlled vacuum theorem.

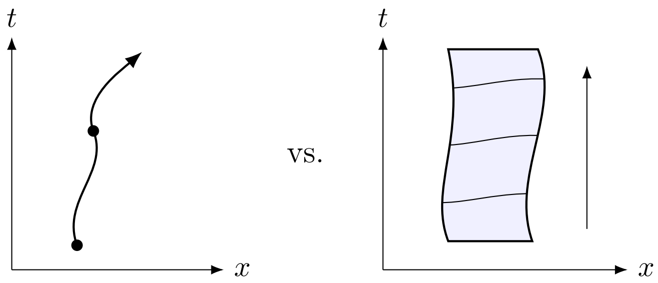
1 Lecture 1: The why and how of Calabi–Yau

Anchor: TASI [7], §§ 1–2.

Guiding question. What can toric methods compute for string compactifications, and where do current methods still fall short?

The answer will be deliberately two-sided. Toric geometry gives an explicit, computable supply of Calabi–Yau threefold hypersurfaces: Hodge numbers, intersection forms, cones, divisor topology, periods, and enumerative inputs such as Gopakumar–Vafa invariants can be extracted algorithmically. It does not, by itself, solve the Ricci-flat metric, determine every effective curve or divisor on the hypersurface, or provide access to all quantum corrections.

Historical and computational thread. The geometric problem starts with two classical inputs. Yau's solution of the Calabi conjecture implies the existence of Ricci-flat Kähler metrics on compact Kähler manifolds with trivial canonical bundle [8], while Calabi–Yau threefolds entered string compactification through the work of Candelas, Horowitz, Strominger and Witten [9]. The reason these lectures focus on toric hypersurfaces is more computational: Batyrev's reflexive polytopes give a large, explicit class of mirror pairs [5]; the Kreuzer–Skarke classification makes this class finite and searchable [6]; and modern packages such as `CYTools` turn the resulting combinatorial



particle theory: worldline

string theory: worldsheet Σ

Figure 1: A point particle sweeps out a one-dimensional worldline, whereas a string sweeps out a two-dimensional worldsheet. In both panels the object evolves upward in target time t: the point traces a curve, while the spatially extended string traces a strip.

data into the intersection, cone and divisor-topology inputs used later [10]. Lecture 1 explains enough of this chain to make Lectures 2 and 3 self-contained.

Goals. After this lecture you can

1. state what a string compactification is, as a geometric object, and why the vacuum Einstein equations on a product ansatz select Ricci-flat internal manifolds;
2. explain why Calabi–Yau threefolds are the computable Ricci-flat class used in these lectures, and how reduced holonomy gives the parallel spinors responsible for residual supersymmetry;
3. identify the moduli of a Calabi–Yau compactification with specific cohomological data;
4. explain why Calabi–Yau hypersurfaces in toric varieties are the natural constructive source of explicit compact examples, and what data they feed into Lectures 2 and 3.

1.1 Why string theory, and why compactify?

Let us motivate how geometry enters physics. We have very successful but structurally different theories for matter and gravity:

$$\begin{array}{lcl} \text{matter} & : & \text{Standard Model,} \\ & & \textit{language:}\ \text{quantum field theory,} \\ & & \textit{hypothesis:}\ \text{point particles;} \\ \text{gravity} & : & \text{General Relativity,} \\ & & \textit{language:}\ \text{differential geometry,} \\ & & \textit{hypothesis:}\ \text{spacetime as a manifold.} \end{array} \tag{1.1}$$

The quantum gravity problem is to find a framework in which these two descriptions can be combined. Perturbative quantum gravity based on point particles is not perturbatively renormalisable, and curvature singularities indicate that the classical spacetime description itself can break down. String theory is one proposed framework in which the fundamental extended objects are one-dimensional strings rather than point particles.

The useful contrast is the following. A particle traces out a worldline in spacetime; a string traces out a worldsheet Σ, and the simplest classical action is proportional to its area,

$$S_{\mathrm{NG}} = -T \int_{\Sigma} \mathrm{d}A\,, \tag{1.2}$$

where T is the string tension. Quantising the string gives a tower of vibrational modes. In the point-particle picture one quantises the possible motions of a point; in the string picture one also quantises the normal modes of an extended object. Very schematically,

$$\text{particle: worldline} \qquad \longrightarrow \qquad \text{string: worldsheet with oscillator modes}\,. \tag{1.3}$$

The word "harmonics" here refers to these oscillator modes of the string. After quantisation they appear, from the spacetime point of view and for an observer sufficiently far away, as different particle species: scalars, gauge fields, spinors, and a spin-2 graviton among them.

It is useful to separate two attitudes. One can use string theory and related constructions as mathematical tools in broader effective descriptions. In these lectures we use the stronger quantum gravity interpretation: the theory is required to include gravity, and the perturbative spectrum contains a spin-2 graviton. With this interpretation, the consistency conditions of the quantum theory are part of the input. In particular, for the critical superstring the consistency of the quantum theory fixes the target spacetime dimension to ten. At energies small compared with the string scale, one keeps only the massless modes. The massive oscillator modes have masses of order the string scale M_s, so a low-energy observer with characteristic energy $E \ll M_s$ cannot excite them as dynamical fields. Their effects are still present indirectly, through higher-derivative corrections to the effective action. The one fact from this construction that Lecture 1 needs is that the massless spectrum of a consistent superstring includes a ten-dimensional metric, together with further geometric fields such as differential forms, scalars, spinors, and gauge fields. The metric is what will make Ricci-flat geometry enter immediately.

Since observed physics is effectively four-dimensional at accessible energies, a compactification must explain how a ten-dimensional theory can give a four-dimensional effective theory:

$$\dim \mathcal{M}_{10} = 10 = 9+1, \qquad \dim \mathcal{M}_4 = 4 = 3+1, \qquad \mathcal{M}_{10} \simeq \mathcal{M}_4 \times X. \tag{1.4}$$

The compact six-dimensional factor X is where geometry enters. The central point of the lectures is that the topology and geometry of X determine the light four-dimensional fields, their couplings, and the scalar potential whose critical values are the vacuum energies.

1.2 Framing: what a string compactification is

For the first lecture, the input from string theory can be stated without choosing a specific string theory. It is the following geometric package.

1. A *ten-dimensional spacetime* $\mathcal{M}_{10}$: a Lorentzian 10-manifold with metric g_{MN}.

2. Additional *fields* Φ on $\mathcal{M}_{10}$. Here a field simply means a geometric object varying over spacetime: a section of a vector bundle, such as a differential form, a function, a connection, a spinor, or the metric itself. In Lecture 1 we keep these fields mostly silent and set them to zero or to constants when deriving the Ricci-flatness condition.

3. *Source terms and charged objects*, such as branes and orientifold planes, which enter through $\mathcal{L}_{\text{source}}$ and produce localised stress-energy and charge. A brane is an extended object on which open strings can end, or more generally an object carrying charge under one of the differential-form gauge fields of the theory. Its worldvolume is a submanifold of spacetime, so after compactification a brane can wrap a cycle in X and fill some or all of the visible four-dimensional spacetime. This is why branes are simultaneously geometric objects and physical sources: they contribute localised energy-momentum, carry quantised charge, and can support gauge fields or matter in the four-dimensional theory [11].

4. An *action functional*

$$S[g,\Phi] = \frac{1}{2\kappa_{10}^2}\int_{\mathcal{M}_{10}} \mathrm{d}^{10}x\,\sqrt{-g}\,\Big(R^{(10)} + \mathcal{L}_{\text{matter}}(g,\Phi) + \mathcal{L}_{\text{source}}(g,\Phi) + \underbrace{\cdots}_{\text{e.g. } R^4}\Big)\,, \tag{1.5}$$

whose Euler–Lagrange equations are the equations of motion, i.e., the differential equations obtained by requiring the first variation of S to vanish, see Eq. (1.9) below. This is the classical, leading-order ten-dimensional action. The full string effective action also contains higher-derivative and quantum corrections.[1] We do not include those corrections in the first-pass vacuum equation, but they reappear in Lecture 2 as corrections to the four-dimensional effective action.

Let x^μ be coordinates on a four-dimensional Lorentzian manifold $\mathcal{M}_4$ and y^m coordinates on a compact real 6-manifold X. A warped product metric on $\mathcal{M}_{10} = \mathcal{M}_4 \times X$ has the form

$$\mathrm{d}s_{10}^2 = \mathrm{e}^{2A(y)}\, g^{(4)}_{\mu\nu}(x)\,\mathrm{d}x^\mu \mathrm{d}x^\nu + \mathrm{e}^{-2A(y)}\, g^{(6)}_{mn}(y)\,\mathrm{d}y^m \mathrm{d}y^n. \tag{1.6}$$

The function $A(y)$ is the *warp factor*. In the first pass through the vacuum equations we set $A = 0$, so the metric is the ordinary product metric $\mathrm{d}s_{10}^2 = \mathrm{d}s^2(\mathcal{M}_4) + \mathrm{d}s^2(X)$. This is the Kaluza–Klein bridge in its simplest form: a four-dimensional observer does not see the internal coordinates directly, but sees the coefficients of expansions in functions, forms, and metric deformations on X.

Definition 1.1 (String compactification, geometrically)**.** *A string compactification consists of: (i) a factorisation $\mathcal{M}_{10} = \mathcal{M}_4 \times X$ of the ten-dimensional spacetime into a four-dimensional Lorentzian manifold $\mathcal{M}_4$ and a compact 6-dimensional Riemannian manifold X (the* internal space*); (ii) a solution of the ten-dimensional equations of motion on $\mathcal{M}_{10}$ with metric in the warped product form (1.6); (iii) background choices for any non-metric fields and charged objects that are present, subject to the global consistency conditions of the chosen string theory.*

On a compact internal space, closed differential-form field strengths can have quantised periods. Abstractly, if F_p is a closed p-form field strength, *flux quantisation* means that its normalised periods over integral p-cycles are integers, so the background defines an integral cohomology class. The precise normalisation and even the correct cohomology theory depend on the string theory and on the charged objects present. Lecture 2 specialises this statement to Type IIB, where the flux superpotential uses the three-form classes $[F_3], [H_3] \in H^3(X, \mathbb{Z})$.

At this point the lecture has only defined the geometric problem: choose a compact six-manifold X, a metric on $\mathcal{M}_4 \times X$, and background fields and sources so that the ten-dimensional equations of motion are satisfied, and then determine the four-dimensional fields and interactions obtained by expanding around that solution. We now first find the internal geometries that can solve the equations of motion in the compactification background, and then ask which four-dimensional fields and potentials those geometries produce.

1.3 The vacuum solution problem

We begin in the simplest setting: no fluxes, no branes. We ask which Riemannian metrics on X are compatible with a vacuum solution.

The metric equation comes from varying the schematic action above with respect to g^{MN}. Let

$$S_{\text{matter+source}} = \frac{1}{2\kappa_{10}^2}\int \mathrm{d}^{10}x\,\sqrt{-g}\,\left(\mathcal{L}_{\text{matter}} + \mathcal{L}_{\text{source}}\right), \tag{1.7}$$

where $\mathcal{L}_{\text{matter}}$ contains the kinetic terms of the non-metric fields Φ and $\mathcal{L}_{\text{source}}$ contains localised brane/orientifold terms. Define

$$T_{MN} := -\frac{2}{\sqrt{-g}}\,\frac{\delta S_{\text{matter+source}}}{\delta g^{MN}}. \tag{1.8}$$

Then the ten-dimensional Einstein equation is

$$R^{(10)}_{MN} = \kappa_{10}^2\left(T_{MN} - \frac{1}{8} g_{MN} T^K{}_K\right). \tag{1.9}$$

[1] The first well-known perturbative correction to the ten-dimensional type-II action contains an R^4 term at order $(\alpha')^3$. The parameter α' is the inverse string tension, up to the conventional factor 2π, and sets the square of the string length. Corrections in powers of α' are higher-derivative or finite-string-size corrections. The string coupling g_s controls the loop expansion.

Holonomy	Dimension	Ricci-flat?	Geometry relevant here
$\mathrm{SO}(n)$	n	no generic implication	no parallel spinor in general
$\mathrm{U}(m)$	$2m$	not necessarily	Kähler
$\mathrm{SU}(m)$	$2m$	yes	Calabi–Yau (Kähler+Ricci-flat)
$\mathrm{Sp}(m)$	$4m$	yes	hyperkähler
$\mathrm{Sp}(m)\cdot\mathrm{Sp}(1)$	$4m$	no (Einstein)	quaternionic Kähler
G_2	7	yes	G_2 manifolds
$\mathrm{Spin}(7)$	8	yes	$\mathrm{Spin}(7)$ manifolds

Table 1: Berger's list of irreducible Riemannian holonomies.

In this first pass we set all additional fields to zero or constant and include no localised sources, so $T_{MN} = 0$. The unwarped product version of (1.6) then gives

$$R^{(4)}_{\mu\nu} = 0\,, \qquad R^{(6)}_{mn} = 0\,. \tag{1.10}$$

The four-dimensional Einstein vacuum equation forces $\mathcal{M}_4$ to be Ricci-flat. For our purposes the physically interesting case is $\mathcal{M}_4 = \mathbb{R}^{1,3}$ (Minkowski), so $R^{(4)}_{\mu\nu} = 0$ holds trivially. The interesting condition is the internal one: X *must be Ricci-flat*.

The problem is then: construct compact Ricci-flat Riemannian 6-manifolds on which one can compute. This is a hard nonlinear PDE problem: explicit Ricci-flat metrics on compact non-flat 6-manifolds are generally not known in closed form. The useful insight is that one does not need the metric explicitly in order to build the compactification. Yau's theorem guarantees the *existence* of Ricci-flat Kähler metrics from complex-geometric data, and toric geometry gives a large explicit class of spaces for which the relevant topological data are computable.

1.4 Berger's holonomy classification

The compactification problem above asks for Ricci-flat internal manifolds with enough structure to control the low-energy theory. Holonomy is the efficient invariant for this purpose. It records how tangent vectors, tensors and spinors transform under parallel transport, and therefore tells us which parallel geometric objects a metric can admit.

Theorem 1.2 (Berger, holonomy classification [12, 13])**.** *Let (X, g) be a simply-connected, complete Riemannian manifold whose holonomy representation is irreducible and whose metric is not locally symmetric. Then the (restricted) holonomy group $\mathrm{Hol}(X, g) \subseteq \mathrm{SO}(\dim X)$ is one of groups in Table 1. The lower bounds $m \geq 2$ remove degenerate or duplicated low-dimensional cases (*$\mathrm{U}(1) = \mathrm{SO}(2)$*,* $\mathrm{Sp}(1) = \mathrm{SU}(2)$*,* $\mathrm{Sp}(1)\cdot\mathrm{Sp}(1) = \mathrm{SO}(4)$*).*

Berger's theorem is useful here because it sharply limits the possible irreducible, non-symmetric Riemannian holonomies. It does *not* classify all compact Ricci-flat manifolds. Rather, it explains why, once one asks for a six-dimensional irreducible Ricci-flat background admitting parallel spinors, the natural holonomy group is contained in $\mathrm{SU}(3)$. In the terminology of [7], the compact Ricci-flat examples used in string compactification all arise from reduced, or special, holonomy; the Calabi–Yau case is the six-dimensional member of this list that is complex, Kähler, and computationally accessible. This is a guide to the known constructive landscape, not a theorem excluding every possible Ricci-flat six-manifold outside the classes we discuss.

For a compact Kähler threefold, the algebraic-geometric condition that implements the $\mathrm{SU}(3)$ holonomy target is the triviality of the canonical bundle. Let $\mathcal{O}_X$ denote the structure sheaf of holomorphic functions on X; an isomorphism $K_X \cong \mathcal{O}_X$ means that the canonical bundle is holomorphically trivial, equivalently that there is a nowhere-vanishing holomorphic $(3, 0)$-form. Holomorphic triviality of K_X gives $c_1(X) = 0$; Yau's theorem then gives a Ricci-flat Kähler metric in each Kähler class; and, after excluding flat or product factors, the holonomy is the expected subgroup of SU(3),

$$K_X \cong \mathcal{O}_X \quad \Longrightarrow \quad \text{Ricci-flat Kähler metric} \quad \Longrightarrow \quad \mathrm{Hol}(X) \subseteq \mathrm{SU}(3)\,. \tag{1.11}$$

When the holonomy is exactly SU(3), this is the generic irreducible Calabi–Yau threefold case. Proper subgroups correspond to extra structure, such as flat factors or product geometries, and usually to more supersymmetry than the generic threefold case. This is the precise sense in which Calabi–Yau threefolds are the workhorse Ricci-flat backgrounds of the lectures: they are explicit enough to construct and compute with, and their reduced holonomy has the spinorial consequence needed by the physics.

1.5 Aside: Reduced holonomy, parallel spinors, and supersymmetry*

Supersymmetry is not the starting point of the geometric argument; it is a consequence of the reduced holonomy just described. The physics reason one cares about it is that residual supersymmetry gives technical control of the low-energy theory and organises the four-dimensional fields into multiplets. The mathematical statement is the existence of parallel sections of the spinor bundle.

Definition 1.3 (Parallel (covariantly constant) spinor)**.** *Let (X, g) be a Riemannian spin manifold with spinor bundle $S \to X$ and let ∇ denote the spin connection. A* parallel *(or* covariantly constant*)* spinor *is a smooth section $\eta \in \Gamma(S)$ with $\nabla\eta = 0$.*

Remark 1.4 (Terminology). In the physics literature these are often called *Killing spinors.* In the standard differential-geometry convention, a *Killing spinor* satisfies $\nabla_X\eta = \lambda\, X\cdot\eta$ for a nonzero constant λ, and the $\lambda = 0$ case is called a *parallel* spinor and treated separately. For the unwarped Minkowski compactification considered here the relevant case is $\lambda = 0$, and we use the term *parallel* throughout.

Proposition 1.5 (Physics dictionary; standard supergravity Killing-spinor analysis, [7] § 2.3)**.** *For the configuration $\mathcal{M}_4 \times X$ with $\mathcal{M}_4 = \mathbb{R}^{1,3}$, the unwarped product metric, no fluxes, and no branes, vanishing of the supergravity SUSY variations on the four-dimensional vacuum requires that X admit a parallel spinor. Starting from a ten-dimensional Type II theory with thirty-two real supercharges, one independent complex parallel spinor on X corresponds to $\mathcal{N} = 2$ four-dimensional supersymmetry (eight real supercharges) before any orientifold projection; starting instead from a heterotic or Type I theory (sixteen real supercharges in ten dimensions), the same condition yields $\mathcal{N} = 1$ in four dimensions (four real supercharges), as in Candelas–Horowitz–Strominger–Witten [9].*

Status: *this is a standard physics derivation, not a mathematical theorem about (X, g) alone, and assumes the leading-order supergravity expansion.*

The invariant internal spinors determine the residual supersymmetry. The precise conversion between internal spinor multiplicities and four-dimensional supercharges depends on the ten-dimensional chirality and real/complex spinor conventions. We only use the standard Type IIB case: an SU(3)-holonomy Calabi–Yau threefold gives $\mathcal{N} = 2$ supersymmetry in four dimensions (eight real supercharges) before the orientifold projection in Lecture 2 reduces it to $\mathcal{N} = 1$.

The bridge to topology is Berger's classification of holonomies on Riemannian manifolds. The curvature condition comes from the commutator of spin covariant derivatives. In an orthonormal frame,

$$\nabla_m\eta = \partial_m\eta + \frac{1}{4}\,\omega_m{}^{ab}\Gamma_{ab}\eta, \qquad [\nabla_m, \nabla_n]\eta = \frac{1}{4}\,R_{mn}{}^{ab}\Gamma_{ab}\eta. \tag{1.12}$$

If η is parallel, the left-hand side vanishes, hence

$$R_{mn}{}^{ab}\Gamma_{ab}\eta = 0. \tag{1.13}$$

By the Ambrose–Singer theorem the restricted holonomy algebra is generated by curvature endomorphisms, so this integrability condition says precisely that the holonomy representation fixes a non-zero spinor. This can happen on a non-flat manifold only if the holonomy group acts trivially on some subspace of the spinor representation; see [14] for the spin geometry and [7], § 2.3 for the physics dictionary.

1.6 Calabi–Yau threefolds: existence and Hodge structure

The six-dimensional geometry problem has now become concrete: find compact Riemannian 6-manifolds whose holonomy is contained in SU(3), and preferably equals SU(3) after excluding torus and product factors. The systematic way to produce such spaces is to construct compact Kähler threefolds with trivial canonical bundle and then apply Yau's theorem to obtain Ricci-flat metrics. Toric Calabi–Yau hypersurfaces are not meant to exhaust the known compact examples with SU(3) holonomy. Rather, Calabi–Yau threefolds are the standard geometric class relevant to this compactification problem, and Batyrev hypersurfaces are the large computable family used in these lectures.

Definition 1.6 (Calabi–Yau threefold)**.** *A* Calabi–Yau threefold *is a compact Kähler manifold X of complex dimension 3 with trivial canonical bundle $K_X \cong \mathcal{O}_X$ and $\pi_1(X) = 0$.*

Equivalently, X admits a nowhere-vanishing holomorphic $(3,0)$-form Ω, unique up to scale. The appendix records the standard comparison with other conventions, including formulations in terms of $c_1(X)$ and holonomy.

The simple-connectedness clause excludes T^6, $K3\times T^2$, abelian threefolds, and CY3s obtained as free quotients by a finite group with nontrivial π_1 (e.g. the heterotic Standard-Model quotients of Donagi–Ovrut [15] and Bouchard–Donagi [16]). It implies $h^{1,0}(X) = 0$, removing spurious vector multiplets from $H^{1,0}$ and giving a clean four-dimensional $\mathcal{N} = 1$ truncation in Lecture 2. The converse fails in general: free finite quotients can have torsion π_1 while still satisfying $h^{1,0} = 0$. The rigorous classification of compact Kähler manifolds with $c_1(X) = 0$ in $H^2(X,\mathbb{R})$ is given by the Beauville–Bogomolov decomposition [17, 12]: such an X admits a finite étale cover that is a product of complex tori, strict Calabi–Yau factors (with $\mathrm{Hol} = \mathrm{SU}(n)$, $n \geq 3$), and irreducible hyperkähler factors (with $\mathrm{Hol} = \mathrm{Sp}(n)$).

Theorem 1.7 (Yau [8])**.** *Let X be a compact Kähler manifold with $c_1(X) = 0$. Then in each Kähler class $[J] \in H^{1,1}(X) \cap H^2(X,\mathbb{R})$ there exists a unique Kähler metric g with $\mathrm{Ric}(g) = 0$.*

Corollary 1.8. *Calabi–Yau threefolds admit Ricci-flat Kähler metrics; in particular they solve the vacuum Einstein equation $R^{(6)}_{mn} = 0$ required by the unwarped product ansatz.*

What follows is the cohomological data we will need in Lecture 2:

Proposition 1.9 (CY3 Hodge diamond)**.** *Let X be a Calabi–Yau threefold. The only independent Hodge numbers are $h^{1,1}$ and $h^{2,1}$, and the Euler characteristic is $\chi(X) = 2\big(h^{1,1} - h^{2,1}\big)$.*

The holomorphic $(3,0)$-form Ω is unique up to overall rescaling and is nowhere vanishing. The pair $(h^{1,1}, h^{2,1})$ is the principal topological invariant of X for our purposes; we will see that $h^{1,1}$ counts *Kähler moduli* (deformations of J) and $h^{2,1}$ counts *complex structure moduli* (infinitesimal deformations of the complex structure on X, equivalently classes in $H^1(X,T_X)$).

Geometric datum	Cohomology group	Dimension
Kähler deformations	$H^{1,1}(X,\mathbb{R})$	$h^{1,1}$
Complex structure deformations	$H^1(X,T_X) \cong H^{2,1}(X)$	$h^{2,1}$
Holomorphic volume form	$H^{3,0}(X)$	1

1.7 Massless spectrum and moduli of CY compactifications

We now identify the four-dimensional moduli with cohomological data on X, following [7] § 2.5.

1.7.1 Geometric moduli: Kähler and complex structure

Proposition 1.10 (Kähler moduli)**.** *The space of infinitesimal Ricci-flat Kähler deformations of (X, g) at fixed complex structure is $H^{1,1}(X, \mathbb{R})$. Concretely, every δJ with $\delta g_{i\bar{j}} = \mathrm{i}\,\delta J_{i\bar{j}}$ harmonic of type $(1, 1)$ deforms the metric to a nearby Ricci-flat one.*

Proposition 1.11 (Complex structure moduli; Kodaira–Spencer)**.** *The space of infinitesimal complex structure deformations of X is $H^1(X, T_X) \cong H^{2,1}(X)$, where the isomorphism is given by contraction with the holomorphic volume form Ω,*

$$\mu \;\mapsto\; \iota_\mu \Omega \;\in\; H^{2,1}(X)\,, \qquad (\iota_\mu\Omega)_{ij\bar{k}} \;=\; \Omega_{ijl}\,\mu^{l}{}_{\bar{k}}\,, \tag{1.14}$$

where $\mu \in H^1(X, T_X)$ is the Beltrami differential, a $(0, 1)$-form valued in $T^{1,0}X$.

We write $J = t^a D_a$, $a = 1, \ldots, h^{1,1}$, for real Kähler coordinates in a chosen divisor basis, with admissibility meaning $J \in \mathcal{K}(X)$, and z^i, $i = 1, \ldots, h^{2,1}$, for complex parameters on $H^{2,1}(X)$. These are the precise geometric realisations of the schematic parameters $u^i(x)$ in §1.2: they are infinitesimal metric deformations δg for which the deformed metric remains Ricci-flat to first order. Yau's theorem upgrades these infinitesimal deformations to genuine nearby Ricci-flat Kähler metrics inside the Kähler cone and complex structure moduli space.

1.7.2 How ten-dimensional fields become four-dimensional fields

We can now turn to the idea of Kaluza–Klein reductions. Suppose first that $A = 0$ and that the internal space X is fixed. A ten-dimensional scalar field admits a formal expansion in eigenfunctions of the scalar Laplacian on X,

$$\Phi(x, y) = \sum_i \varphi_i(x) f_i(y), \qquad -\Delta_X f_i = \lambda_i f_i. \tag{1.15}$$

After inserting this expansion into the ten-dimensional kinetic term and integrating over X, the coefficient $\varphi_i(x)$ becomes a four-dimensional scalar field with mass squared proportional to λ_i, up to the conventional length scale of X and to corrections from warping or background fluxes. Thus harmonic modes, $\lambda_i = 0$, give massless fields before additional interactions are included, while non-zero eigenvalues give the Kaluza–Klein tower. A field is called *light* if its mass lies below the cutoff of the four-dimensional effective theory under consideration.

The same idea applies to differential forms. If C_p is a p-form field and ω_α is a harmonic p-form on X, then

$$C_p(x, y) = \sum_\alpha a_\alpha(x)\,\omega_\alpha(y) + \cdots\,, \qquad [\omega_\alpha] \in H^p(X, \mathbb{R}), \tag{1.16}$$

produces scalar fields $a_\alpha(x)$ in four dimensions. This is the first reason cohomology appears in the physics: it organises the massless topological modes.

1.8 Energy densities from moduli fields

The previous subsection explained how cohomology classes and families of internal metrics give rise to light four-dimensional fields. The next question is dynamical: once these fields have been identified, what energy is assigned to a configuration of them? This is the point at which the geometry of the compact space becomes an input to four-dimensional cosmology.

For the physics, the crucial extra step is to allow these parameters to depend on the four-dimensional coordinates x^μ. A path in the moduli space of Ricci-flat metrics then becomes a four-dimensional field configuration. If the fields vary in time, the internal metric $g^{(6)}(y; \phi(t))$ also changes in time, and this time-dependence contributes to the four-dimensional stress-energy.

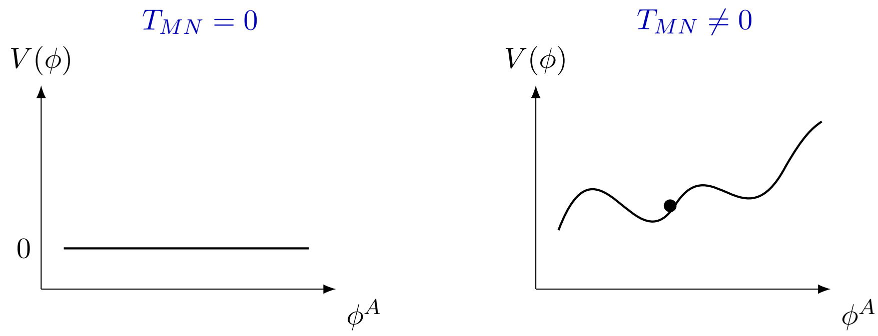


Figure 2: Classically the Calabi–Yau moduli are flat directions: in the leading source-free approximation the ten-dimensional stress-energy contribution is $T_{MN} = 0$, and the four-dimensional potential vanishes along the moduli space. Fluxes, localised sources, and further corrections are represented schematically by $T_{MN} \neq 0$ in the ten-dimensional equations; in the four-dimensional theory they lift the flat moduli space to an energy landscape $V(\phi)$.

In this sense, the variation of the geometry of X leads directly to cosmological evolution in $3+1$ dimensions.

The phrase *cosmological evolution* will be used here in a minimal sense. It means a solution of the four-dimensional Einstein equations coupled to scalar fields, typically with a time-dependent metric on $\mathcal{M}_4$ and time-dependent scalar fields $\phi^A(t)$. From the ten-dimensional point of view, this same solution can be read as a family of internal geometries $g^{(6)}(y; \phi(t))$ evolving over four-dimensional time. Thus the dynamical evolution of the internal metric is not an extra metaphor: it is precisely the four-dimensional dynamics of the moduli fields.

This brings the discussion back to the central aim of the lecture series. Moving in the moduli space of internal geometries is, in the four-dimensional effective theory, the same as changing the values of scalar fields. The scalar potential on this field space is the object whose critical values are the vacuum energies that Lectures 2 and 3 compute from toric data.

The metric itself also reduces. A family of Ricci-flat metrics $g^{(6)}(y; \phi)$ depending on parameters ϕ^A gives four-dimensional scalar fields by allowing those parameters to vary over spacetime,

$$g^{(6)}_{mn}(y; \phi) \quad \rightsquigarrow \quad g^{(6)}_{mn}\big(y; \phi(x)\big). \tag{1.17}$$

If varying ϕ^A preserves the leading-order ten-dimensional equations, then ϕ^A is a *modulus*. For Calabi–Yau compactifications, the most important examples are the Kähler moduli and complex structure moduli counted by $h^{1,1}(X)$ and $h^{2,1}(X)$. They are geometric fields in four dimensions: their values determine the shape and size data of the internal Calabi–Yau metric. Classically, these moduli have no potential: moving in the moduli space still gives a vacuum solution, so

$$V(\phi) = 0 \qquad \text{along the classical Calabi–Yau moduli space.} \tag{1.18}$$

Later we will include ingredients with non-trivial stress-energy, such as fluxes, branes, orientifold planes, and quantum corrections. In ten dimensions these appear as $T_{MN} \neq 0$ in (1.9); in the four-dimensional effective theory they generate a scalar potential $V(\phi)$ for the fields ϕ^A. At the level of the resulting four-dimensional effective action one has schematically

$$S_4 = \int_{\mathcal{M}_4} \mathrm{d}^4x \sqrt{-g_4} \left[\frac{M_{\mathrm{Pl}}^2}{2} R^{(4)} - G_{AB}(\phi)\, \partial_\mu \phi^A \partial^\mu \phi^B - V(\phi) + \cdots \right]. \tag{1.19}$$

For constant scalar fields, the scalar equations of motion reduce to

$$\partial_A V(\phi_*) = 0. \tag{1.20}$$

The four-dimensional Einstein equation then relates the value of the potential to the spacetime curvature,

$$R^{(4)}_{\mu\nu} = \frac{V(\phi_*)}{M^2_{\rm Pl}}\, g^{(4)}_{\mu\nu} \quad \text{in these conventions.} \tag{1.21}$$

This is the energy-landscape picture used throughout the lectures: geometry supplies fields and functions; a vacuum is a critical point; and its vacuum energy is the critical value $V(\phi_*)$. Lecture 2 identifies the geometric and arithmetic data needed to write the four-dimensional functions K, W, and hence V; Lecture 3 then uses these functions to compute $V(\phi_*)$ at explicit candidate critical points.

1.9 Why Calabi–Yau hypersurfaces in toric varieties?

The logic above tells us which manifolds are admissible; it does not tell us which ones we can compute on. The practical compactification programme is to enumerate large families of Calabi–Yau threefold topologies, compute the cohomological and intersection-theoretic data controlling the four-dimensional fields, compute periods and instanton corrections where needed, and then use these inputs to study potentials and vacuum energies. For computability we use the toric construction of Batyrev [5], with which the audience is already familiar. Projective hypersurfaces such as the quintic are familiar first examples; Batyrev hypersurfaces in toric varieties are the high-dimensional computational generalisation. A one-paragraph summary, for the sake of fixing notation:

Theorem 1.12 (Batyrev; CY threefolds from reflexive polytopes)**.** *Let $\Delta^\circ \subset N_{\mathbb{R}}$ be a reflexive 4-polytope (so $\Delta = (\Delta^\circ)^\vee \subset M_{\mathbb{R}}$ is also a lattice polytope, and Δ° has the origin as its unique interior lattice point). Let $\Sigma_{\mathcal{T}}$ be the fan of a fine, regular, star triangulation $\mathcal{T}$ of Δ°. Then $V_{\Sigma_{\mathcal{T}}}$ is an MPCP toric ambient variety: it is allowed to have terminal quotient singularities, but its singular locus does not meet a generic anticanonical hypersurface in the threefold case. Consequently the generic anticanonical hypersurface $X = \{f = 0\} \subset V_{\Sigma_{\mathcal{T}}}$ is a smooth Calabi–Yau threefold, and Batyrev gives explicit combinatorial formulas for its Hodge numbers in terms of Δ, Δ° alone.*

The Kreuzer–Skarke list [6] of all Δ° contains 473 800 776 reflexive 4-polytopes; together with the FRSTs $\mathcal{T}$ this gives a landscape of explicit Calabi–Yau hypersurfaces in toric varieties covering $1 \le h^{1,1} \le 491$. This is the input to Lectures 2 and 3: every concrete construction will be “on a particular Batyrev hypersurface in a specified toric ambient.”

Here is the toric punchline in the form needed for the rest of the lectures. The Batyrev input $(\Delta^\circ, \mathcal{T})$ is not the final object of interest. It is the starting point for a chain of computations

$$(\Delta^\circ, \mathcal{T}) \rightsquigarrow X \rightsquigarrow \{\text{topological data, cones, periods, etc.}\} \rightsquigarrow (K, W, V) \rightsquigarrow V(\phi_*)\,. \tag{1.22}$$

The last number is the vacuum energy of a four-dimensional critical point. The list below is therefore a dictionary, not a collection of unrelated toric facts. For each entry one should keep three questions separate: what is the geometric object, where does it enter the four-dimensional calculation, and how is it obtained from the toric description?

- *Wall data.* A torus-invariant divisor $\widehat{D}_\rho$ on the ambient toric variety restricts to a divisor $D_\rho = \widehat{D}_\rho \cap X$ on X, possibly empty for lattice points interior to facets. In favourable cases these restricted toric divisor classes span $H^{1,1}(X)$. Choosing a basis D_a, one computes

$$\kappa_{abc} = \int_X D_a \wedge D_b \wedge D_c, \qquad c_{2,a} = \int_X c_2(X) \wedge D_a. \tag{1.23}$$

Together with the Hodge numbers $h^{1,1}, h^{2,1}$, this completes the *Wall data* of a Calabi–Yau threefold. For $J = t^a D_a$, these same numbers give the Calabi–Yau volume and divisor volumes

$$\begin{aligned} \mathcal{V}(X,J) &= \frac{1}{3!}\int_X J\wedge J\wedge J = \frac{1}{6}\kappa_{abc}t^a t^b t^c, \\ \tau_a &= \frac{1}{2}\int_{D_a} J\wedge J = \frac{1}{2}\int_X D_a\wedge J\wedge J = \frac{\partial\mathcal{V}}{\partial t^a} = \frac{1}{2}\kappa_{abc}t^b t^c. \end{aligned} \tag{1.24}$$

Used in: the Kähler potential, the mirror prepotential, and the non-perturbative superpotential.

How: In practice this data is obtained from the ambient toric intersection ring, the hypersurface class, and the Chern-class formula for toric hypersurfaces. The implementation used in the computational parts of these lectures is `CYTools` [10]; the same data are the Wall data used to compare toric phases in [18].

- *Cone data and flops.* The Kähler cone $\mathcal{K}(X)$ records which real $(1,1)$-classes can be represented by Kähler forms; its dual Mori cone records effective curve classes. Curve volumes are the inequalities that keep the large-volume expansion under control. Different FRSTs can give adjacent Kähler chambers related by flops, and the union of these chambers is the extended Kähler cone $\widetilde{\mathcal{K}}(X)$. In Lecture 3, moving through these adjacent chambers is the geometric part of the vacuum search: one follows the potential while checking that all relevant curve volumes remain positive and sufficiently large.

  Used in: the domain of validity of the large-volume expansion, the definition of Kähler chambers, and the geometric transitions used in the Lecture 3 search strategy.

  How: `CYTools` computes Mori and Kähler cone data for toric phases and tracks the classical data across triangulations [10]. The secondary-fan and chamber-navigation layer used in the separate Lecture 3 and 4 discussion is provided by `fanroots` [19].

- *Divisor data.* Non-perturbative terms in the superpotential may come from ED3 instantons or gaugino condensation on divisors. A necessary zero-mode test is expressed in terms of the Hodge numbers of the divisor, especially $h^{0,1}(D)$ and $h^{0,2}(D)$, or equivalently the arithmetic genus appearing in Witten's criterion. For prime toric divisors these Hodge numbers can often be read from the face of Δ dual to the lattice point defining the divisor. This is why a purely combinatorial computation can screen candidate instanton divisors before any four-dimensional potential is written.

  Used in: deciding which divisors can support D-instanton terms in W_{np}.

  How: For prime toric divisors, compute divisor Hodge numbers via the face-duality formulae, see e.g. [20]. This is a special toric shortcut. For non-inherited or non-toric divisors, determining the relevant cohomology groups is a substantially harder sheaf-cohomology problem and should not be treated as automatic output of the toric combinatorics.

- *Period data.* The flux superpotential in Type IIB is built from periods of the holomorphic three-form,

  $$\Pi(z) = \left(\int_{\Gamma_I}\Omega(z)\right), \qquad \Gamma_I \in H_3(X,\mathbb{Z}). \tag{1.25}$$

  Near a large complex structure point these periods are computed by mirror symmetry. In practice this means solving the Picard–Fuchs system of the mirror polytope and expressing the answer in flat coordinates. These periods determine W_{flux}, and hence the flux value W_0 after the complex structure moduli and the axio-dilaton are stabilised.

  Used in: the flux superpotential W_{flux}, in particular the stabilisation of complex structure moduli.

How: construct the GKZ / Picard–Fuchs system from the Batyrev mirror data, compute the Frobenius solutions near an LCS point, and then match the logarithmic solutions to an integral symplectic period basis using the large volume monodromy on the mirror side [21, 22, 23].

- *Enumerative data.* Genus-zero Gopakumar–Vafa invariants n_β, for $\beta \in H_2(X, \mathbb{Z})$, are the integer invariants that repackage genus-zero Gromov–Witten invariants after the multiple-cover contributions have been separated [24, 23]. For the audience it is useful to remember two interpretations. Mathematically, they are the integral BPS/GV refinement of rational curve counts in class β. In the physical interpretation they are indexed BPS degeneracies, naturally described by M2-branes wrapping holomorphic curves in class β. They appear in the worldsheet-instanton part of the prepotential, so they correct the periods and can generate exponentially small terms used in small-W_0 constructions. Lecture 2 is devoted to making this computational mirror symmetry (CMS) / GV calculation explicit enough for the vacuum energy story.

  Used in: worldsheet-instanton corrections to the prepotential, convergence estimates for the instanton expansion, and the small terms that can enter small-W_0 and conifold-PFV constructions.

  How: compare the instanton part of the mirror prepotential with the Frobenius period expansion, use the multiple-cover formula to pass from genus-zero Gromov–Witten coefficients to integral GV invariants, and organise the resulting finite computation by Mori-cone degree or by a decomposition-closed truncation [24, 21, 23]. In Lecture 3 the same curve-class data are then compared across flop-related chambers.

Many of the conceptual ingredients in this story are classical by now: Calabi–Yau compactification, mirror symmetry, periods, flux superpotentials, instanton corrections, and Gopakumar–Vafa invariants have been studied for decades. What has changed rather dramatically is our ability to compute these ingredients in practice, not only for a few hand-picked examples, but uniformly across large families with widely varying Hodge numbers. The main breakthrough for explicit string constructions is therefore algorithmic as much as conceptual: toric geometry, mirror symmetry, triangulation algorithms, period computations, and modern searches over flux and Kähler data turn the existence of a vast landscape into concrete, checkable Diophantine problems.

The companion `CYTools` session demonstrates the computational pipelines on explicit examples: querying the Kreuzer–Skarke database, constructing or sampling FRSTs, building a `CalabiYau` object, computing intersection numbers and cone data, testing divisor topology, and extracting truncated GV dictionaries. The remainder of these lectures explains why these computations are the inputs to established string theory constructions.

Lecture 1 punchlines. The first lecture should leave the following takeaways. First, compactification turns internal geometry into four-dimensional fields; the light fields are controlled by cohomology, harmonic forms, and Ricci-flat metric deformations. Second, uncorrected Calabi–Yau compactifications have flat moduli spaces, so vacuum energies only become isolated numbers after non-trivial sources of stress-energy generate a potential. Third, Calabi–Yau threefolds enter because Yau's theorem turns the nonlinear Ricci-flat metric problem into a computable complex-geometric existence problem. Fourth, hypersurfaces a la Batyrev make this computable at scale: toric data provide the intersection, cone, divisor, period, and GV inputs that Lectures 2 and 3 feed into the four-dimensional scalar potential.

That completes Lecture 1.

A General conventions and notation

We collect the conventions used throughout. This is a reference section for the written notes, not a preliminary lecture. Each main lecture repeats the minimum notation and equations it needs. The conventions follow [7], § 2.2, and have been streamlined for the present audience.

A.1 Indices and spacetime

The ten dimensions of the ambient string-theoretic spacetime are indexed by capital Latin letters $M, N, \ldots = 0, \ldots, 9$. The ansatz of physical interest splits these into four "external" dimensions $\mu, \nu, \ldots = 0, 1, 2, 3$ corresponding to the four-dimensional spacetime that our cosmology lives in, and six "internal" dimensions $m, n, \ldots = 4, \ldots, 9$ corresponding to a compact real 6-manifold X:

$$\mathrm{d}s^2_{10} \;=\; \mathrm{e}^{2A(y)}\, g^{(4)}_{\mu\nu}(x)\, \mathrm{d}x^\mu \mathrm{d}x^\nu \;+\; \mathrm{e}^{-2A(y)}\, g^{(6)}_{mn}(y)\, \mathrm{d}y^m \mathrm{d}y^n\,. \tag{A.1}$$

The factor $\mathrm{e}^{2A(y)}$ is the *warp factor*; we will set it to unity at first and reinstate it in Lecture 3. We use signature $(-, +, +, \ldots, +)$ for $g^{(10)}$.

A.2 Differential forms and the Hodge decomposition

For a real C^∞ manifold M of dimension D we write $\Omega^r(M)$ for the space of smooth r-forms, $\mathrm{d} : \Omega^r \to \Omega^{r+1}$ for the exterior derivative, and $\star : \Omega^r \to \Omega^{D-r}$ for the Hodge star with respect to a Riemannian metric g. The adjoint $\mathrm{d}^\dagger = (-1)^{Dr+D+1} \star \mathrm{d} \star$ and the Laplacian $\Delta = \mathrm{d}\mathrm{d}^\dagger + \mathrm{d}^\dagger \mathrm{d}$ act on $\Omega^r(M)$.

We denote by $\mathcal{H}^r(M)$ the space of harmonic r-forms. For M compact orientable, Hodge's theorem identifies

$$H^r(M, \mathbb{R}) \;\cong\; \mathcal{H}^r(M)\,. \tag{A.2}$$

We write $b_r(M) = \dim_{\mathbb{R}} H^r(M, \mathbb{R})$ for the Betti numbers.

A.3 Complex manifolds and Dolbeault cohomology

A *complex manifold* of complex dimension k is a real $2k$-manifold with complex coordinate charts whose transition functions are holomorphic. The tangent bundle splits $TM \otimes \mathbb{C} = T^{1,0}M \oplus T^{0,1}M$, and differential forms decompose into (p, q)-types,

$$\Omega^r(M, \mathbb{C}) \;=\; \bigoplus_{p+q=r} \Omega^{p,q}(M)\,, \tag{A.3}$$

with $d = \partial + \bar\partial$ on a Hermitian manifold and $\bar\partial^2 = 0$. The $\bar\partial$-Laplacian $\Delta_{\bar\partial} = \bar\partial\bar\partial^\dagger + \bar\partial^\dagger\bar\partial$ defines harmonic (p, q)-forms; we write $\mathcal{H}^{p,q}_{\bar\partial}(M)$ for the space and $h^{p,q}(M) = \dim \mathcal{H}^{p,q}_{\bar\partial}(M)$ for the *Hodge numbers*.

On a Kähler manifold (Hermitian metric whose Kähler form J is closed) one has the identity $\Delta_d = 2\Delta_\partial = 2\Delta_{\bar\partial}$, hence

$$b_r(M) \;=\; \sum_{p+q=r} h^{p,q}(M)\,, \qquad h^{p,q} \;=\; h^{q,p}\,, \qquad h^{p,q} \;=\; h^{k-p,k-q}\,. \tag{A.4}$$

A.4 Calabi–Yau threefolds

There is more than one convention for "Calabi–Yau"; we record ours explicitly. *In these notes a Calabi–Yau n-fold is a compact Kähler manifold X of complex dimension n with trivial canonical bundle $K_X \cong \mathcal{O}_X$ and $\pi_1(X) = 0$.* The convention matches that used in the candidate-dS literature [7, 1]; the simple-connectedness clause is convenient but not standard across the differential-geometry literature (Joyce, for instance, defines a CY n-fold as a compact Kähler manifold with $\mathrm{Hol}(X) = \mathrm{SU}(n)$, which implies $\pi_1(X)$ finite but not necessarily trivial). Under our convention, the following standard structural facts hold; in each case the implication should be read in the indicated direction.

- Triviality of K_X is equivalent to the existence of a nowhere-vanishing holomorphic $(n,0)$-form $\Omega \in H^0(X, K_X) = H^{n,0}(X)$.
- Yau's theorem (Thm. 1.7 below) gives the existence and uniqueness of a Ricci-flat Kähler metric in each Kähler class.
- For this Ricci-flat metric, the restricted holonomy (which, under $\pi_1(X) = 0$, equals the full holonomy) satisfies $\mathrm{Hol}(X) \subseteq \mathrm{SU}(n)$; equality $\mathrm{Hol}(X) = \mathrm{SU}(n)$ requires further input, namely that X is irreducible (not a Riemannian product) and not hyperkähler, in which case Berger's classification (Theorem 1.2) gives $\mathrm{Hol}(X) = \mathrm{SU}(n)$.

For $n = 3$ the Hodge diamond and the resulting Euler characteristic $\chi(X) = 2\big(h^{1,1}(X) - h^{2,1}(X)\big)$ are stated in §1.6 (Proposition 'CY3 Hodge diamond'); we do not repeat them here.

A.5 Toric notation (reminder; not derived)

We use the standard toric-geometry conventions of [4]. For a complete simplicial fan $\Sigma \subset N_{\mathbb{R}}$ (with $N \cong \mathbb{Z}^d$) we write V_Σ for the associated toric variety. Prime toric divisors D_ρ correspond to rays $\rho \in \Sigma(1)$. The Chow group $A^1(V_\Sigma) \otimes \mathbb{R}$ has a distinguished cone $\mathcal{K}(V_\Sigma)$ (the Kähler cone) dual to the Mori cone $\mathcal{M}(V_\Sigma) := \overline{\mathrm{NE}}(V_\Sigma)$ (the closure of the cone generated by classes of effective 1-cycles; see Definition **??**). The notation $\mathcal{M}$ is used in the lectures to avoid confusing the Mori cone with the nef cone; the standard algebraic-geometry notation is displayed at the point of definition. For a 4-dimensional reflexive polytope $\Delta^\circ \subset N_{\mathbb{R}}$ (with polar dual $\Delta \subset M_{\mathbb{R}}$, $M = N^\vee$), the Batyrev construction [5] produces a Calabi–Yau threefold X as the generic anticanonical hypersurface in the ambient toric fourfold $V_{\Sigma_{\mathcal{T}}}$ associated to a fine, regular, star triangulation (FRST) $\mathcal{T}$ of Δ° (Appendix **??**). The ambient $V_{\Sigma_{\mathcal{T}}}$ has at worst $\mathbb{Q}$-factorial terminal Gorenstein singularities, located in cones of codimension ≥ 4 in $V_{\Sigma_{\mathcal{T}}}$ (i.e. at isolated points), which generically do not meet the smooth hypersurface X.

We write $\kappa_{abc} = D_a \cdot D_b \cdot D_c$ for the triple intersection numbers of (pullbacks of) prime toric divisors on X. For the Calabi–Yau hypersurface X, the Kähler cone is denoted $\mathcal{K}(X) \subset H^{1,1}(X, \mathbb{R})$; the *extended Kähler cone* $\widetilde{\mathcal{K}}(X)$ is the union of $\mathcal{K}(X')$ over all birational models X' reached by chains of flops, glued along their common flop walls (see Proposition **??** and Appendix **??**).

The mirror Calabi–Yau threefold of X is denoted $\widetilde{X}$ (Batyrev construction, Appendix **??**); we do not use the X° notation. Note that the tilde has two unrelated uses in these notes: the mirror manifold $\widetilde{X}$ and the extended Kähler cone $\widetilde{\mathcal{K}} = \widetilde{\mathcal{K}}$.

A.6 Conventions for field-theory quantities

We work in $\hbar = c = 1$ units, and adopt $M_{\mathrm{Pl}} = 1$, where the four-dimensional reduced Planck mass has dimensionful value $M_{\mathrm{Pl}} \simeq 2.4 \times 10^{18}$ GeV. A *field* on a spacetime manifold is a geometric object allowed to vary from point to point: for example a function, a differential form, a connection, or a metric. An *action* is a functional of these fields,

$$S[\Phi, g] = \int_M \mathrm{d}^{\dim M} x \, \sqrt{|g|}\, \mathcal{L}(\Phi, g, \nabla\Phi, \ldots), \tag{A.5}$$

where Φ denotes all non-metric fields and $\mathcal{L}$ is the Lagrangian density. Varying S with respect to Φ gives the field equations. These field equations are what physicists call the *equations of motion*: the differential equations satisfied by classical field configurations that are stationary points of the action. Varying S with respect to the metric gives the stress-energy tensor

$$T_{MN} := -\frac{2}{\sqrt{-g}} \frac{\delta S_{\text{matter+source}}}{\delta g^{MN}} \,. \tag{A.6}$$

Thus a schematic ten-dimensional action of the form

$$S^{(10)} = \frac{1}{2\kappa_{10}^2} \int \mathrm{d}^{10}x \, \sqrt{-g} \left(R^{(10)} + \mathcal{L}_{\text{matter}} + \mathcal{L}_{\text{source}} \right) \tag{A.7}$$

gives the Einstein equation

$$R^{(10)}_{MN} - \frac{1}{2} g_{MN} R^{(10)} = \kappa_{10}^2\, T_{MN}. \tag{A.8}$$

Taking the trace in 10 dimensions rewrites this as $R^{(10)}_{MN} = \kappa_{10}^2 (T_{MN} - \frac{1}{8} g_{MN} T)$, with $T = T^K{}_K$.

A four-dimensional field is called *light* relative to an effective description with cutoff Λ if its mass is well below Λ, so it is retained as a dynamical variable. Fields with masses comparable to or above Λ are integrated out and affect the light theory only through corrections to its couplings. In a compactification, harmonic zero modes are massless at leading order and hence light before fluxes, branes, or quantum corrections generate a potential.

The ten-dimensional gravitational coupling κ_{10}^2 is normalised so that the leading-order action takes the form

$$S^{(10)} = \frac{1}{2\kappa_{10}^2} \int \mathrm{d}^{10}x\, \sqrt{-g_{10}} \left(R^{(10)} - \tfrac{1}{2}|\partial\tau|_\tau^2 - \tfrac{1}{2}|G_3|^2 - \tfrac{1}{4}|\widetilde{F}_5|^2 + \dots \right), \tag{A.9}$$

with $\tau = C_0 + \mathrm{i}e^{-\phi}$ the axio-dilaton, $G_3 = F_3 - \tau H_3$ the complexified three-form, and $\widetilde{F}_5$ the self-dual five-form (precise form: (**??**)). The norm $|\partial\tau|_\tau^2$ uses the hyperbolic metric on the upper half-plane.

Remark A.1 (A note on physics terms). Throughout these lectures, "supergravity" (SUGRA) denotes the low-energy field-theory approximation to a chosen string background: an action-based field theory on a Lorentzian manifold of fixed dimension, retaining the massless fields and their leading interactions. "Supersymmetry" (SUSY) denotes a $\mathbb{Z}_2$-graded extension of the Poincaré algebra by spinorial generators; an $\mathcal{N} = 1$ theory in D dimensions has one such spinor generator (so 4 real supercharges in $D = 4$). "Flux" denotes a properly quantised closed differential form, or more precisely its integral cohomology class. In the supersymmetric geometric approximation, a "brane" is represented by a calibrated submanifold of X carrying gauge / matter degrees of freedom. Precise definitions appear in Lectures 2 and 3.

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